



05506026

Digital Image Processing

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CONTENTS



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SPATIAL FILTERING



SPATIAL FILTERING

Filter

- ▶ Filter is a sub-image which contains co-efficient values in each pixel
- ▶ Filter usually contains small odd number of pixels. For example, 3x3, 5x5, 7x7
- ▶ They are also well-known in other names
 - ▶ Kernel
 - ▶ Mask
 - ▶ Template
 - ▶ Window

Mechanics of Spatial Filter

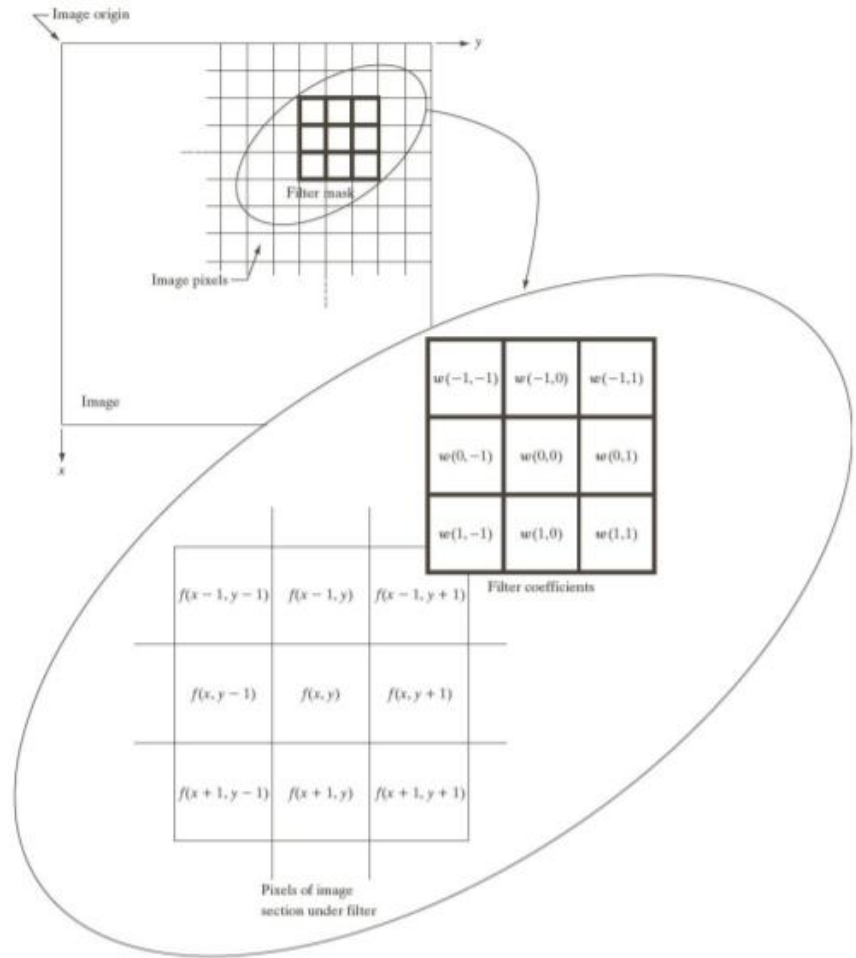
- ▶ Spatial filter consists of
 - ▶ A neighbourhood
 - ▶ Predefined operation that performed on the image pixels surrounded by the neighbourhood
- ▶ Filtering operation is to create a new pixel with coordinates equal to the centre of neighbourhood
- ▶ If the operation is linear, the filter is called linear spatial filter
- ▶ If the operation is not linear, the filter is called nonlinear spatial filter

- For example, a linear spatial filtering using 3x3 neighbourhood will be calculated using

$$\begin{aligned}
 g(x, y) &= w(-1, -1)f(x - 1, y - 1) \\
 &+ w(-1, 0)f(x - 1, y) + \dots + w(0, 0)f(x, y) \\
 &+ \dots w(1, 1)f(x + 1, y + 1)
 \end{aligned}$$

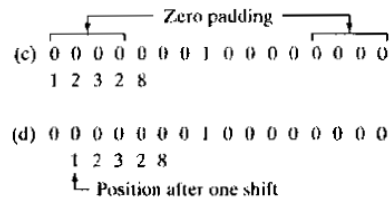
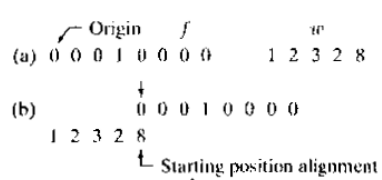
Or

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t)f(x + s, y + t)$$



Mechanics of Spatial Filter

- ▶ In case that the filter is out of bound of the image, there are some solutions for this problem
 - ▶ Assign the value 0 to pixel in the out-of-bound area, which called “zero padding”



- ▶ Remove that area



Smoothing (Lowpass) Spatial Filters

Smoothing Spatial Filters

- ▶ There are two purposes for this filter
 - ▶ Blur the image
 - ▶ Reduce the noise in the image

Smoothing Linear Filters

- ▶ The output of smoothing linear filter is the average of the pixels contained in the neighbourhood of the filter mask, sometimes called “averaging filters” or “lowpass filters” (filter which pass low frequency to the image)
- ▶ The result image will be reduced in part of sharp transition, for example, noise, also edges

Smoothing Linear Filters

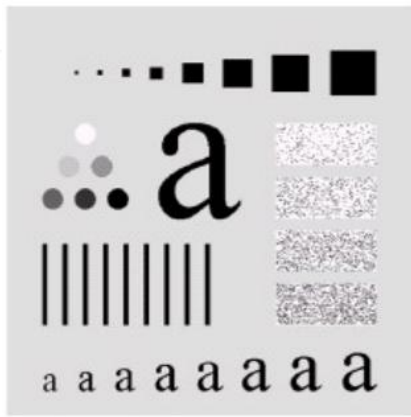
- For example, if we use 3x3 mask, we can do simple standard average of the pixels as follow

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

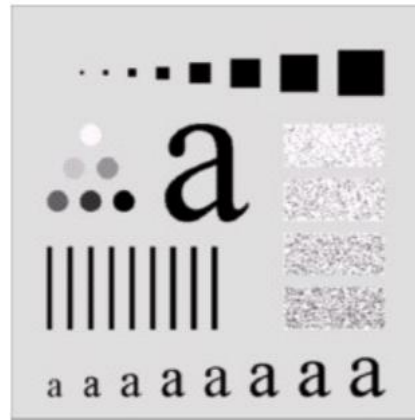
Smoothing Linear Filters

- ▶ But if we need to focus the importance of some pixels, we can perform weighed average filter instead

$$\frac{1}{16} \times \begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline \end{array}$$



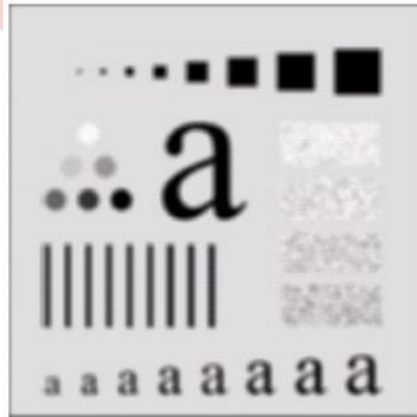
Original Image



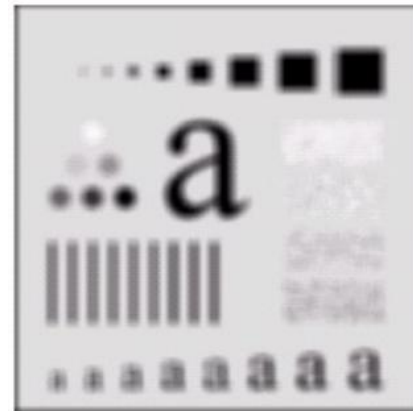
Filtered with 3x3



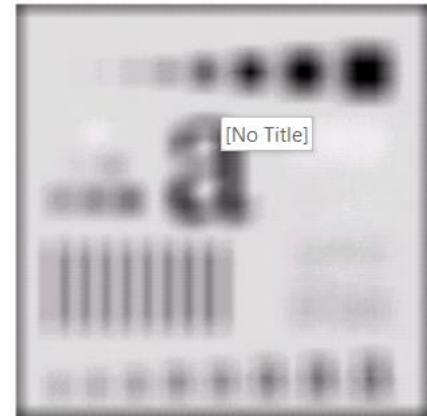
Filtered with 5x5



Filtered with 9x9

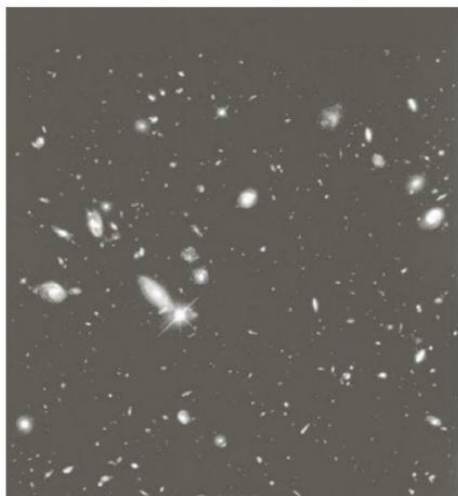


Filtered with 15x15

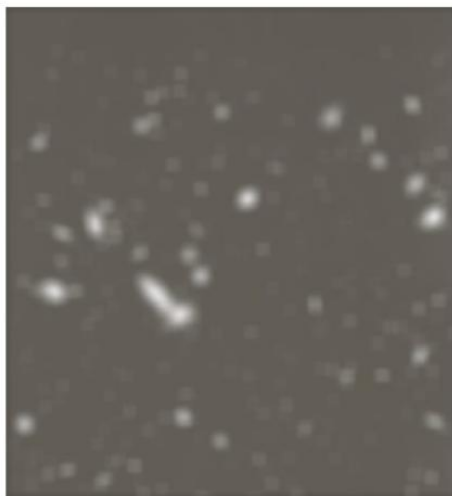


Filtered with 35x35

Smoothing Linear Filters



Original Image
(528 x 485 pixels
from the Hubble Space Telescope)



Filtered with 15x15



Thresholding

Order-Statistic (Nonlinear) Filters

- ▶ Order-statistic filters are nonlinear spatial filters whose response is based on ordering (ranking) the pixels contained in the image are surrounded by the filter, then replacing the value of the centre pixel with the value determined by the ranking
- ▶ The best-known filter is “median filter”
- ▶ Excellent for noise reduction especially “impulse noise”, or so called “salt-and-pepper noise” (white and black noise)

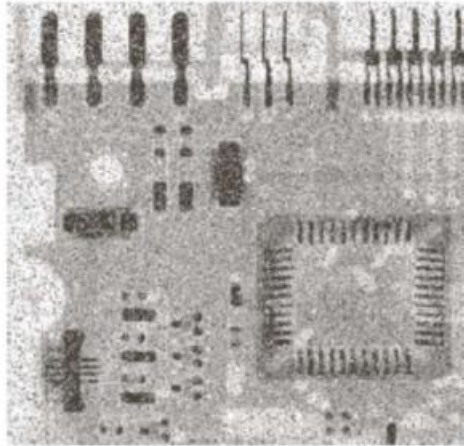
Order-Statistic (Nonlinear) Filters

- ▶ Median filter

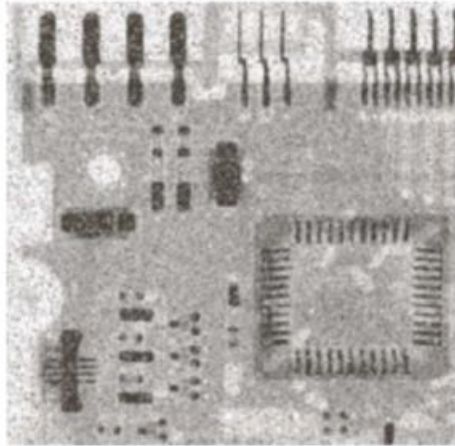
	1	2	3
	7	8	9
	4	5	6

- ▶ The result of the middle pixel will be 5 (the median)

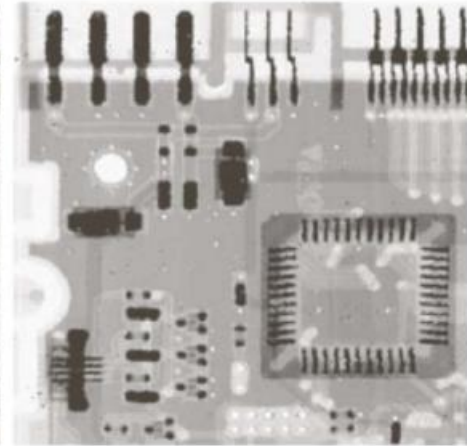
Order-Statistic (Nonlinear) Filters



**Original Image
(Salt-and-pepper noise)**



**Filtered with 3x3
averaging mask**



**Filtered with 3x3
median mask**



Sharpening (Highpass) Spatial Filters

Sharpening Spatial Filters

- ▶ Used to sharpen the image
- ▶ Based on first and second order derivatives

- ▶ First derivative

$$\frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

x

-1	1
----	---

- ▶ Second derivative

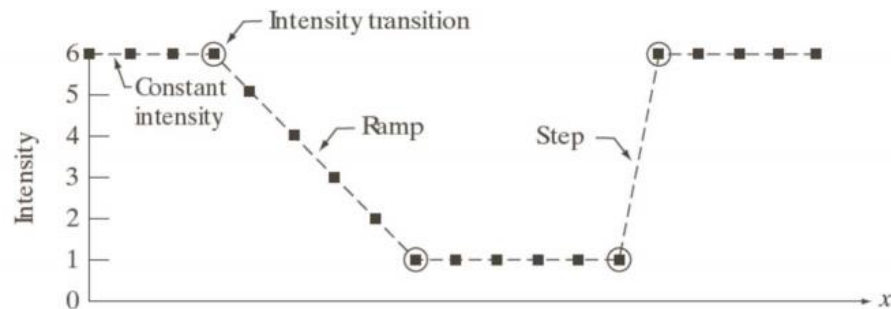
$$\frac{\partial^2 f}{\partial x^2} = f(x + 1) + f(x - 1) - 2f(x)$$

x

1	-2	1
---	----	---

Sharpening Spatial Filters

- ▶ We will consider image in 3 parts
 - ▶ Flat segment (constant intensity)
 - ▶ Step
 - ▶ Ramp

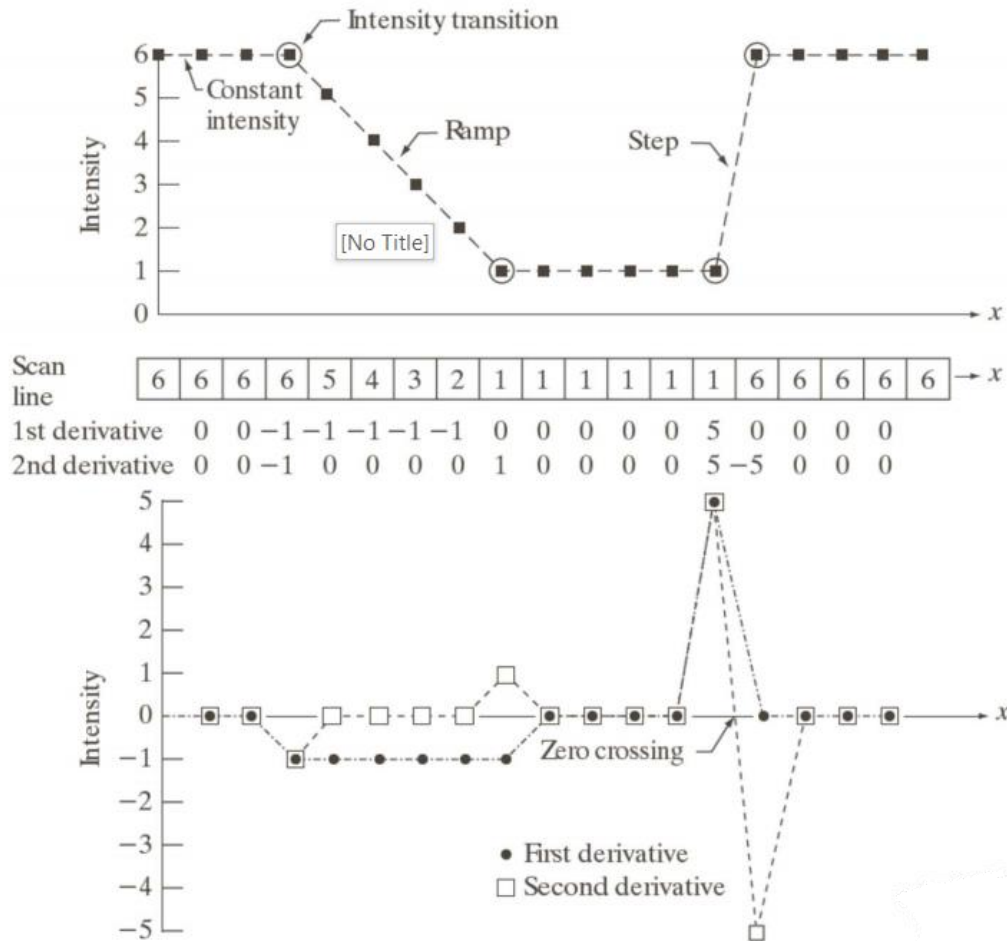


Sharpening Spatial Filters

- ▶ When we apply first derivative: -
 - ▶ In the area of constant intensity, the value must be ZERO
 - ▶ At the onset of an intensity step or ramp, the value must be NONZERO
 - ▶ Along the ramp, the value must be NONZERO

Sharpening Spatial Filters

- ▶ When we apply second derivative: -
 - ▶ In the area of constant intensity, the value must be ZERO
 - ▶ At the onset AND at the end of an intensity step or ramp, the value must be NONZERO
 - ▶ Along the ramp WITH CONSTANT SLOPE, the value must be ZERO



The Laplacian

(using the second derivative for image sharpening)

- ▶ The Laplace operator or Laplacian is a differential operator which is given by the divergence of the gradient of a function on Euclidean space
- ▶ In Cartesian coordinates, the Laplacian is given by

$$\nabla^2 f = \frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2}$$

The Laplacian

(using the second derivative for image sharpening)

- ▶ Laplacian can be derived into a discrete form in x-direction and y-direction using second derivative as follow

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

Therefore,

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

The Laplacian

(using the second derivative for image sharpening)

- Therefore, Laplacian mask is

0	1	0
1	-4	1
0	1	0

The Laplacian

(using the second derivative for image sharpening)

- In real practice,

Original Laplacian

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

Laplacian for diagonal

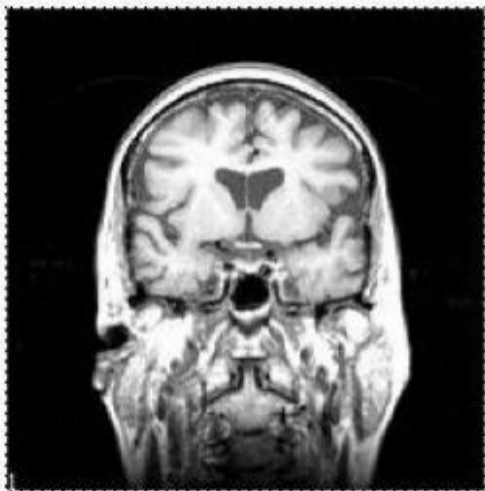
Alternative Laplacian

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

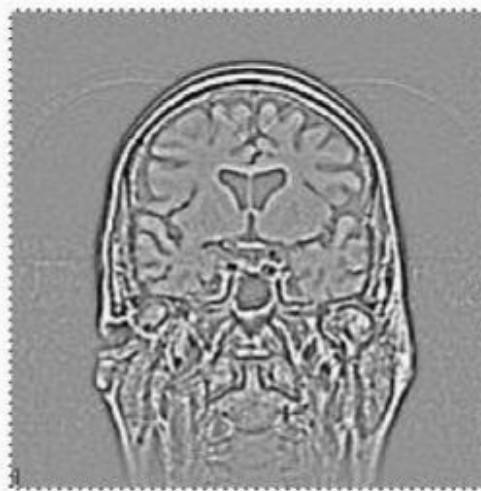
Alternative Laplacian for diagonal

The Laplacian

(using the second derivative for image sharpening)



(A) Original MR image



(B) Laplacian results



(C) Extraction of the zero crossing of the Laplacian (object edges)

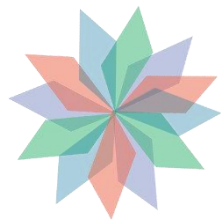
The Laplacian

(using the second derivative for image sharpening)





Unsharp Masking



Unsharp masking and Highboost filtering

- ▶ Unsharp marking is a process to sharpen the image using a lot in printing and publishing industry
- ▶ The process is to build a mask using smoothed version of the image and subtract it from the original image
- ▶ The process is: -
 - ▶ Blur the original image
 - ▶ Subtract the blurred image from the original image (get a mask)
 - ▶ Add mask with the original image

Unsharp masking and Highboost filtering

- ▶ Blur the original image

$$\bar{f}(x, y)$$

- ▶ Subtract the blurred image from the original image (get a mask)

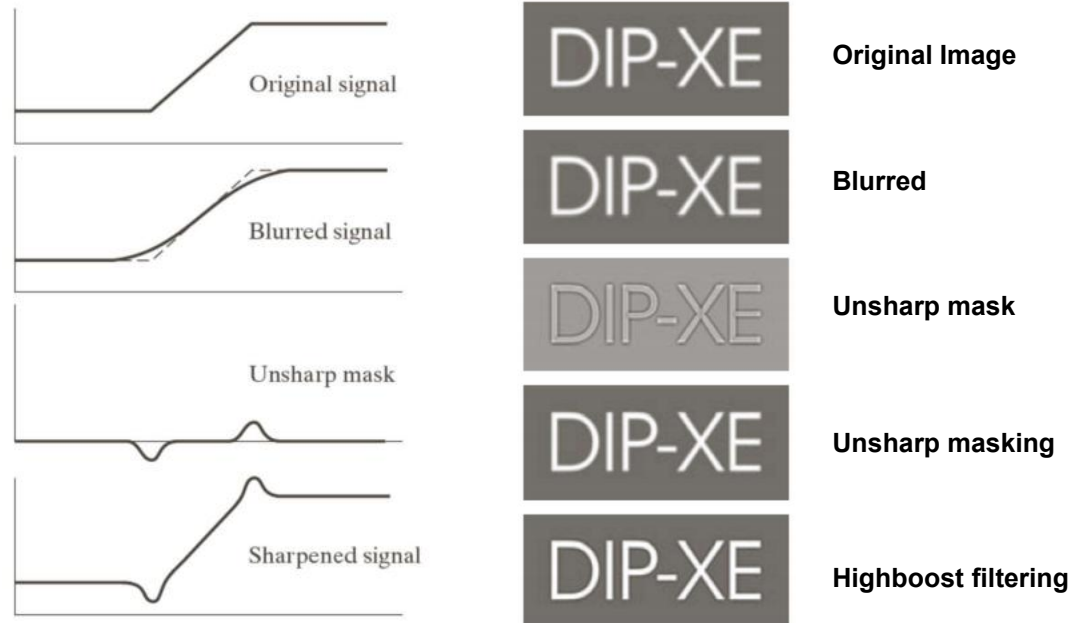
$$g_{mask}(x, y) = f(x, y) - \bar{f}(x, y)$$

- ▶ Add mask with the original image

$$g(x, y) = f(x, y) + k \times g_{mask}(x, y)$$

- ▶ If $k = 1$, we call Unsharp masking
- ▶ If $k > 1$, we call Highboost filtering

Unsharp masking and Highboost filtering



Unsharp masking and Highboost filtering

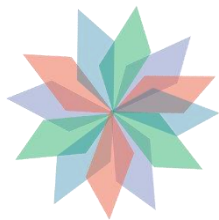


Original Image

Slight unsharp masking Strong unsharp masking



Gradient



The Gradient

(using the first derivative for nonlinear image sharpening)

- ▶ The Gradient of f at coordinate (x,y) is defined as the two-dimensional column **vector**

$$\nabla f = \text{grad}(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

The Gradient

(using the first derivative for nonlinear image sharpening)

- ▶ If we denote a 3x3 image as

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

- ▶ z_5 is the centre or $f(x,y)$

The Gradient

(using the first derivative for nonlinear image sharpening)

- ▶ First derivative in x-direction (Approx.) is

$$\frac{\partial f(x+1)}{\partial x} = f(x+1) - f(x) \rightarrow g_{x+1} = (z_6 - z_5)$$

$$\frac{\partial f(x)}{\partial x} = f(x) - f(x-1) \rightarrow g_x = (z_5 - z_4)$$

The Gradient

(using the first derivative for nonlinear image sharpening)

- ▶ First derivative in x-direction (Approx.) is

$$\frac{\partial f(x+1)}{\partial x} + \frac{\partial f(x)}{\partial x} = (z_6 - z_5) + (z_5 - z_4) = z_6 - z_4$$

- ▶ If we take weighed constant into the account, then

$$\frac{\partial f}{\partial x} \rightarrow g_x = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

The Gradient

(using the first derivative for nonlinear image sharpening)

- ▶ First derivative in x-direction (Approx.) is

$$\frac{\partial f}{\partial x} \rightarrow g_x = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

- ▶ Therefore, in y-direction

$$\frac{\partial f}{\partial y} \rightarrow g_y = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel operator Y

Sobel operator X

The Gradient

(using the first derivative for nonlinear image sharpening)

- ▶ The magnitude (length) of gradient is

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

- ▶ Or we can use the approximate result as

$$M(x, y) \approx |g_x| + |g_y|$$

The Gradient

(using the first derivative for nonlinear image sharpening)

- ▶ Approximate magnitude is

$$\begin{aligned} M(x, y) &\approx |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)| \\ &\quad + |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| \end{aligned}$$

The Gradient

(using the first derivative for nonlinear image sharpening)

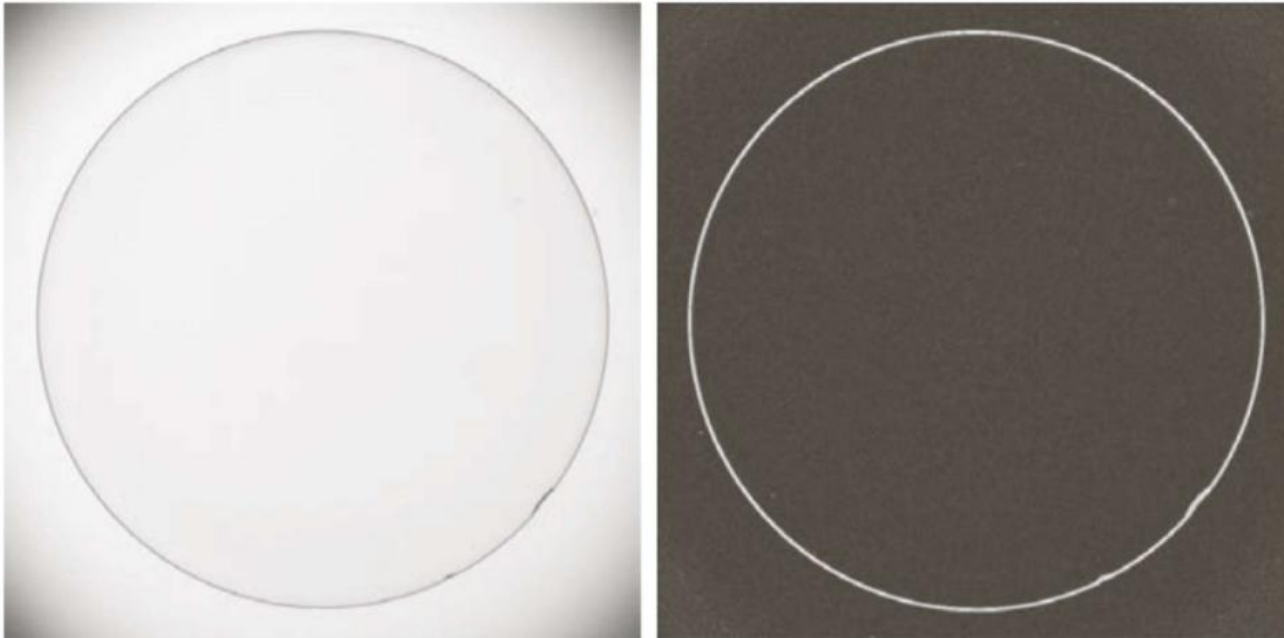
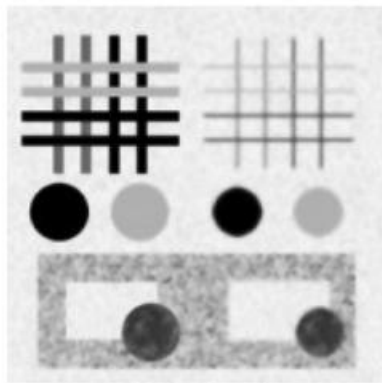
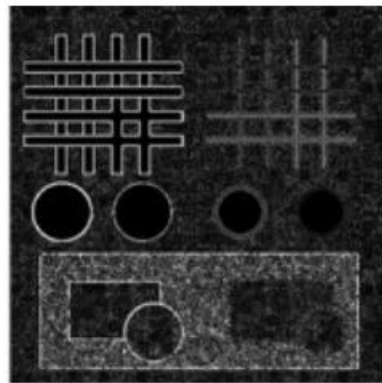


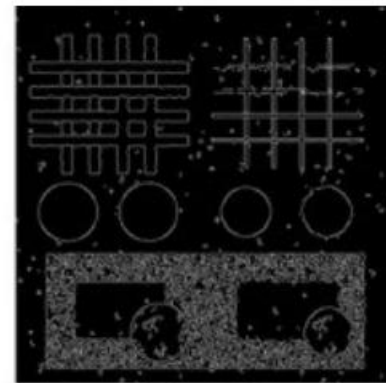
Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.



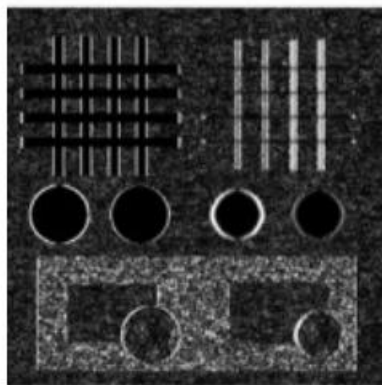
Original



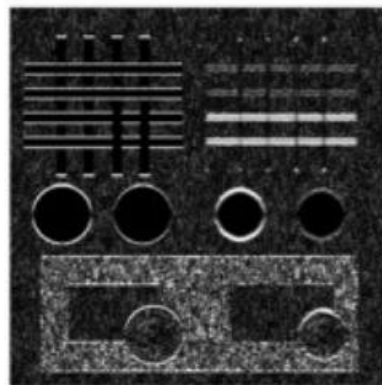
Laplacian



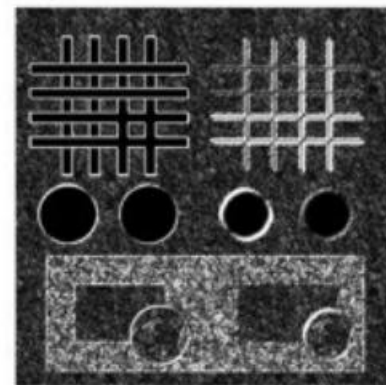
Canny



Sobel X



Sobel Y



Sobel X+Y



End of Week 4